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Expression analysis of ABA- and GA-related genes during four stages of bud dormancy in Japanese apricot (*Prunus mume* Sieb. et Zucc)

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ABSTRACT

Seasonal dormancy in deciduous fruit trees is a complex physiological state, which is regulated by many plant hormones and genes. Abscisic acid (ABA) and gibberellic acid (GA_3) are known to be particularly important in controlling dormancy. We measured the concentrations of these two hormones in flower buds (collected during para-, endo-, and eco-dormancy, and at the dormancy-release stage) and vegetative buds (collected monthly from 20 November 2012 to 20 February 2013) of two cultivars of Japanese apricot (*Prunus mume* Sieb. et Zucc), 'Bungo' and 'Taoxingmei', using liquid chromatography and tandem mass spectrometry (LC–MS/MS). Levels of expression of two zinc-finger family genes *CCCH* and *C2H2* and the *GA20ox* (gibberellin 20 oxidase) gene (GenBank Accession No. Pm013607), as candidates for dormancy-controlling genes, were analysed at each sampling time using real-time RT-quantitative PCR (RT-qPCR). Expression analyses indicated that both zinc-finger family genes exhibited high transcript levels in flower buds of both cultivars during para-dormancy, concomitant with high ABA concentrations. The zinc-finger family genes were relatively highly expressed in vegetative buds of both cultivars on 20 November 2012 (autumn). In addition, high concentration of GA_3 in flower buds at the dormancy-release stage suggested that GA_3 plays an important role in controlling the release of bud dormancy. High levels of expression of the *GA20ox* gene during eco-dormancy and at the dormancy-release stage might have a prominent effect on dormancy-release by regulating gibberellin (GA) signalling in Japanese apricot. These results will contribute to increasing our understanding of ABA- and GA-mediated seasonal dormancy mechanisms in Japanese apricot.

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Prunus mume; ABA; GA; bud dormancy; Japanese apricot

Japanese apricot (*Prunus mume* Sieb. et Zucc) is a deciduous fruit tree that originated in China and has been widely cultivated in Asia for approximately 3,000 years (Chu, 1999). It possesses a short chilling requirement and is the earliest fruit tree species to bloom in the family Rosaceae, making it an appropriate target for research into the mechanisms of woody plant dormancy.

Dormancy is commonly divided into para-, endo-, and eco-dormancy (Lang et al. 1987). Dormancy-release is also an important stage in the growth cycle of woody plants. The timing of the onset of dormancy and dormancy-release in temperate horticultural and perennial forestry species is mediated by at least three regulatory mechanisms involving signal perception, processing by phytochromes (PHY), and PHY-interacting transcription factors (Shim et al., 2014). Abscisic acid (ABA) is involved in regulating the onset of dormancy and in maintaining the dormant state (Karssen et al., 1983). In addition, some specific zinc-finger family proteins can enhance dormancy or, conversely, release dormancy when their genes are down-regulated (Barrero et al., 2010). A previous study suggested that a novel zinc-finger protein, with a proline-rich N-terminus, was an important downstream component of the ABA-signalling pathway that mediated ABA-regulated seed dormancy in *Arabidopsis thaliana* (He & Gan, 2004). Other reports (Feurtado et al., 2011; Bogamuwa & Jang, 2013) have also shown that genes involved in ABA responses are related to zinc-finger protein genes and have roles in seed dormancy and seed germination.

Gibberellins (GAs) are a family of compounds based on the *ent*-gibberellane structure. Only a few GAs show intrinsic biological activity, including GA_1, GA_4, GA_5, GA_6, GA_7, and the most widely available compound, GA_3. GA_3 is used to promote seed germination, stem elongation, and fruit growth in a variety of agronomically and horticulturally important plants (Davies, 2010), and GA_3 was selected for investigation in this study. Gibberellins such as GA_3 might function in timing of establishment of dormancy and in chilling-induced dormancy-release (Schrader et al., 2004). Recent studies have suggested that bud-burst and shoot elongation depend on

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sufficient levels of GAs, such as GA_1, GA_3, and GA_4, and on reductions in the amounts of bio-active GAs rather than on reduced sensitivity to GA, which plays a major role in the cessation of growth (Hoffman, 2011). Barros et al. (2012) suggested that GA metabolism in almond trees might result from the differential regulation of at least the *PdGA20OX* and *PdGA2OX* genes following bud-break before flowering. Lower levels of active GA, due to over-expression of the gene for the catabolic enzyme *GA2-oxidase*, or to impaired GA perception by over-expression of *GA INSENSITIVE* or *RGA-like* receptor genes, resulted in the early onset of dormancy and late bud-burst in poplar trees (Zawaski et al., 2011).

Although many relevant studies have been performed, limited knowledge exists regarding the roles of GA_3, ABA, and their related genes in the control of bud dormancy in Japanese apricot genotypes having different chilling requirements. This study aimed to resolve the mechanisms of ABA- and GA-mediated changes in dormancy of Japanese apricot, using liquid chromatography and tandem mass spectrometry (LC–MS/MS) and real-time RT-quantitative PCR (RT-qPCR), during the key stages of dormancy. The concentrations of both hormones and the patterns of expression of their associated genes were measured, to increase our knowledge of dormancy in woody plants.

Materials and methods

Plant material

Two Japanese apricot (*Prunus mume* Sieb. et Zucc) cultivars, 'Bungo' (high chilling requirement) and 'Taoxingmei' (low chilling requirement), were used. Both were grown at the National Field Genebank for Japanese apricot, located at Nanjing Agricultural University. Flower buds were collected according to the number of chilling hours experienced and phenological traits (Gao et al., 2012). Before leaf loss (approx. early-November, in autumn), flower buds were at the para-dormancy stage. After leaf loss (approx. late-December, in winter), the branches were placed in a phytotron in 5% (w/v) sucrose and were maintained at 25 ± 1°C under white fluorescent light (intensity of 55 μmol m^{-2} s^{-1}) with a 16 h photoperiod, and at 18 ± 1°C for 8 h of darkness, with a constant relative humidity of 70%. The sucrose solution was changed every 3 d, and the basal branches were re-cut at the same time. After 10 d incubation, buds showing green sepal tips were considered to have burst. Flower buds that were less than 50% open, more than 50% open in the phytotron, or 50% open in the field, were considered to be at the stages of endo- or eco-dormancy or dormancy-release (Campoy et al., 2012), respectively. The final sampling dates for 'Bungo' were 10 December 2012, and 11 January, 8 February, and 26 February 2013. The final sampling dates for 'Taoxingmei' were 9 November 2012, and 23 January, 12 December, and 24 February 2013.

Vegetative buds of both cultivars were collected monthly from 20 November 2012 to 20 February 2013, and young leaves (n = 3) were collected on 20 March 2013 from each cultivar.

Total RNA extraction

Total RNA was extracted from 100 mg of frozen flower buds (collected during the stages of para-, endo-, and eco-dormancy, and at dormancy-release), vegetative buds (collected monthly from 20 November 2012 to 20 February 2013), and young leaves of each Japanese apricot tree using the cetyltrimethyl ammonium bromide method (Pavy et al., 2008) with modifications. Genomic DNA contamination was removed using RNase-free DNase I (TaKaRa, Dalian, P. R. China). Total RNA from Japanese apricot flower or vegetative buds and leaves was frozen in liquid nitrogen and stored at −70°C for use.

Expression analysis of GA-related genes and zinc-finger protein-related genes

PCR primers (Table 1) were designed according to published gene sequences (Zhang et al., 2012) using Primer 5.0 software (http://www.premierbiosoft.com/). Real-time RT-PCR (RT-qPCR) was carried out using an Applied Biosystems 7300 Real Time PCR System (Applied Biosystems, Singapore) in 20 μl reaction volumes, each containing 1.0 μl of ten-fold diluted cDNA, 0.2 μl (4 pmol) of each PCR primer, 10.0 μl SYBR® Premix Ex Taq™ (TaKaRa), and 9.1 μl of sterile double-distilled water. The thermal cycling profile used was 95°C for 3 min, then 40 cycles at 95°C for 25 s, 62°C for 25 s, and 72°C for 40 s. The quantity of transcripts of each gene in each sample was normalised using *ACTIN* gene transcripts. Relative levels of expression were calculated using the $2^{-\Delta\Delta CT}$ method (Livak and Schmittgen, 2001).

Reagents and standards

All solvents were of HPLC-grade. Methanol (Merck, Darmstadt, Germany), acetonitrile (Merck), dichloromethane (GC-grade), acetic acid (HPLC-grade), and distilled water (Wahaha, Hangzhou, P. R. China) were used. The ABA and GA_3 standards were purchased from Sigma Aldrich (St. Louis, MO, USA), and were prepared by dissolving them to 1.0 mg ml^{-1} in 100%

Table 1. Japanese apricot genes selected for real-time RT-qPCR analysis and the primers used.

Gene name	GenBank Accession numbers	Product	Forward primer (5′ → 3′)	Reverse primer (5′ → 3′)
CCCH	Pm025964	CCCH-type zinc-finger family protein	TCAACTGGTGTAGGAAGCAAATCG	GCAGGAAGTGGCAACTCTCG
C2H2	Pm004490	C2H2-type zinc-finger family protein	GCTCCTGCCGCATCTTCCTC	CTATTGCTGCTCTGCCCTACTCC
GA20ox	Pm013607	Gibberellin 20 oxidase	AGGATGGTCAGTGGATTGATGTGG	AGGAGAAGAATGCCAGGGAATACC

(v/v) methanol. Benzoic acid (Balex a.s., Pardubice, Czech Republic) was used as an internal standard.

Extraction and purification procedures

Samples of lower bud and vegetative bud tissues (100 mg) were ground to a fine powder in liquid nitrogen, and extracted at 4°C for 12h in 20 ml of 80% (v/v) methanol (Durgbanshi et al., 2005). Before extraction was performed, 10 µg of the internal standard (benzoic acid) was added. The extract was then centrifuged at 1,000 × *g* at 4°C for 15 min, and the supernatant was collected. The extraction was repeated. The pooled supernatants were purified using a solid-phase Agela Cleanert SPE-NH2 extraction column (Agela, Wilmington, DE, USA), dried under N_2, re-suspended in 1.5 ml of 0.05% (v/v) acetic acid in an 85:15 (v/v) mix of H_2O and acetonitrile, and finally filtered through a 0.45 µm Millex-HV filter (Millipore, Bedford, MA, USA).

The recovery rates of ABA and GA_3 were based on the peak area ratios obtained by extraction of standards. Standards of both plant hormones (ABA and GA_3), each at two different concentrations (50 ng ml^{-1} and 100 ng ml^{-1}), were added to each sample before extraction was performed.

Liquid chromatography and tandem mass spectroscopy procedures

A Finnigan LC–MS/MS system (Thermo-Electron, San Jose, CA, USA), consisting of a Surveyor autosampler, a Surveyor MS pump, and a Finnigan LTQ linear ion-trap mass spectrometer equipped with an ESI source, was used

Liquid chromatography (LC) was carried out using an Agilent 1100 quaternary pump (Agilent, Waldrom, Germany). A Supelco Discovery C18 (Supelco, Bellefonte, PA, USA) column (2.1 mm × 150 mm; 5 µm) was used at ambient temperature. The injection volume was 10 µl. An elution gradient was applied using a binary solvent system consisting of 0.05% (v/v) acetic acid in H_2O (solvent A) and 100% (v/v) acetonitrile (solvent B) at a constant flow rate of 400 µl min^{-1} and at a split ratio of 1:3. The linear gradient profile is shown in Table 2. The LC–MS/MS conditions were as follows: ESI spray voltage, 4 kV; sheath gas flow-rate, 70 arbitrary units (arb. units); auxiliary gas flow-rate, 20 arb. units; capillary voltage, −38 V; capillary temperature, 350°C; and offset voltage of the tube lens, 95 V. The selected-reaction monitoring (SRM) mode was used to measure the two phytohormones. ABA and GA_3 concentrations were monitored at *m*/*z* transitions of 263 → 153 and 345 → 143. The optimised collision energy for both ABA and GA_3 was 20 eV. The selected ion monitoring (SIM) mode was used to measure benzoic acid. Benzoic acid was monitored at an *m*/*z* ratio of 121. Preliminary analyses were carried out using a scan from *m*/*z* values of 0–500. ABA was eluted at 4.85 ± 0.03 min, GA_3 was eluted at 2.95 ± 0.03 min, and benzoic acid was eluted at 4.35 ± 0.03 min.

Table 2. Linear gradient elution programme for separating ABA and GA_3 using LC–MS/MS.

Flow rate (ml min^{-1})	Time (min)	Acetonitrile [% (v/v)]	0.05% (v/v) Acetic acid [% (v/v)]
0.4	0.0	20	80
0.4	2.5	20	80
0.4	4.5	98	2
0.4	10.5	98	2
0.4	12.0	20	80

Statistical analysis

Statistical analysis was performed using SPSS software Version 18.0 (SPSS-IBM Inc., Chicago, IL, USA). One-way analysis of variance (ANOVA) was applied to detect differences in hormone concentrations and levels of gene expression between the equivalent stages of dormancy in 'Bungo' and 'Taoxingmei'. Significant differences between the different time-points of bud or leaf samples from each cultivar were also tested (data not shown). Mean differences at $P \leq 0.05$ were considered to be significant using Duncan's multiple range test. All values shown are means ± SD ($n = 3$).

Results

Concentrations of ABA and GA_3 in flower buds

The recovery rates of ABA and GA_3 were calculated based on the 50 ng ml^{-1} and 100 ng ml^{-1} standards added. The mean rates of recovery of ABA were 96.6% and 97.5%, respectively, and of GA_3 were 95.2% and 96.3%, respectively.

ABA concentrations declined in flower buds of 'Bungo' and 'Taoxingmei' during endo-dormancy

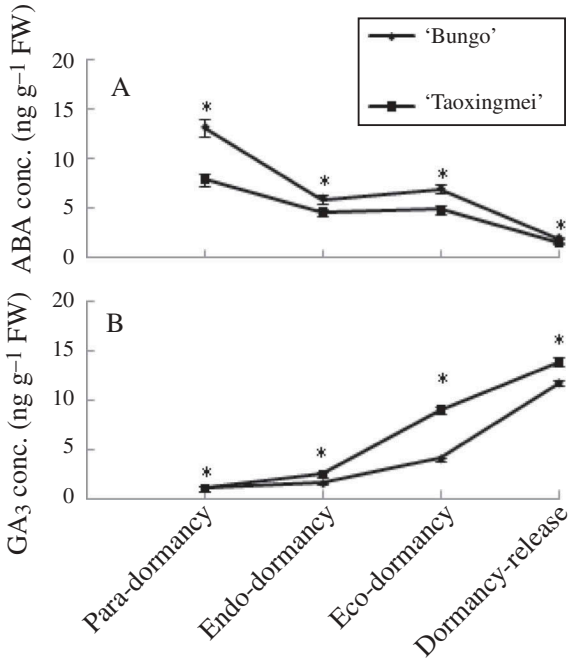


Figure 1. Variations in abscisic acid (ABA) and gibberellic acid (GA_3) concentrations during four key stages of dormancy in flower buds of Japanese apricot 'Bungo' and 'Taoxingmei'. Para-, endo-, and eco-dormancy, and the dormancy-release stages coincided with sampling dates of 10 December 2012, 11 January, 8 February, and 26 February 2013 for 'Bungo' (high chilling requirement), and sampling dates of 9 November and 12 December 2012, and 23 January and 24 February 2013 for 'Taoxingmei' (low chilling requirement). Values are means ± SD (n = 3). Hormone concentrations at each stage of dormancy marked with an asterisk were significantly different between 'Bungo' and 'Taoxingmei' at $P \leq 0.05$ using Duncan's multiple range test.

compared to para-dormancy, then remained constant during eco-dormancy (Figure 1A). A significant reduction in ABA concentration was observed at the dormancy-release stage.

The concentration of GA_3 increased slightly in flower buds of 'Bungo' and 'Taoxingmei' at endo-dormancy compared to para-dormancy, then increased significantly during eco-dormancy (Figure 1B). The concentration of GA_3 was highest at the dormancy-release stage.

During the four key stages of dormancy, the ABA concentration was highest during para-dormancy, and lowest at the dormancy-release stage, in both cultivars. Conversely, the lowest concentration of GA_3 was observed during para-dormancy, and the highest at the dormancy-release stage, in both cultivars. ABA concentrations in flower buds of 'Bungo' were higher than at the corresponding stages of dormancy in 'Taoxingmei', and concentrations of GA_3 in flower buds of 'Taoxingmei' were higher than in flower buds of 'Bungo' at all four key stages of dormancy.

Concentrations of GA_3 in vegetative buds of Japanese apricot

In both cultivars, concentrations of GA_3 (Figure 2) were low on 20 November and 20 December, and higher on 20 January and 20 February. Concentrations of GA_3 in 'Bungo' were noticeably lower than in 'Taoxingmei' from 20 November to 20 February.

Concentrations of GA_3 were low in both cultivars on 20 November and 20 December, and highest on 20 February.

Expression patterns of the zinc-finger family genes CCCH and C2H2 in flower buds

Levels of expression of the zinc-finger family genes *CCCH* (Figure 3A) and *C2H2* (Figure 3B) were significantly lower in flower buds of both cultivars during endo-dormancy than during para-dormancy, and continued to be down-regulated during eco-dormancy, before being highly up-regulated at the dormancy-release stage.

Fewer transcripts of the zinc-finger family genes *CCCH* and *C2H2* were observed in flower buds of 'Taoxingmei' than in 'Bungo' during the para-, endo-, and eco- dormancy, and dormancy-release stages.

Expression patterns of zinc-finger family genes CCCH and C2H2 in vegetative buds and young leaves

Levels of expression of the zinc-finger family genes *CCCH* (Figure 3C) and *C2H2* (Figure 3D) showed a similar trend in vegetative buds of the two cultivars. Transcription of both genes was lower on 20 December than on 20 November, but was higher on 20 January, before being down-regulated in February. Expression levels continued to decline significantly in young leaves of both cultivars. Almost no *C2H2* transcripts were detected in young leaves of 'Taoxingmei'.

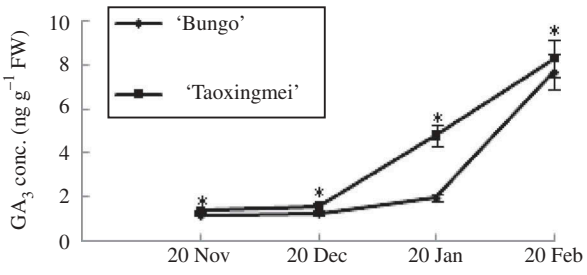


Figure 2. GA_3 concentrations in vegetative buds of two Japanese apricot cultivars, 'Bungo' and 'Taoxingmei'. Values are means ± SD (n = 3). Concentrations at each stage of dormancy marked with an asterisk were significantly different between 'Bungo' and 'Taoxingmei' at $P \leq 0.05$ using Duncan's multiple range test.

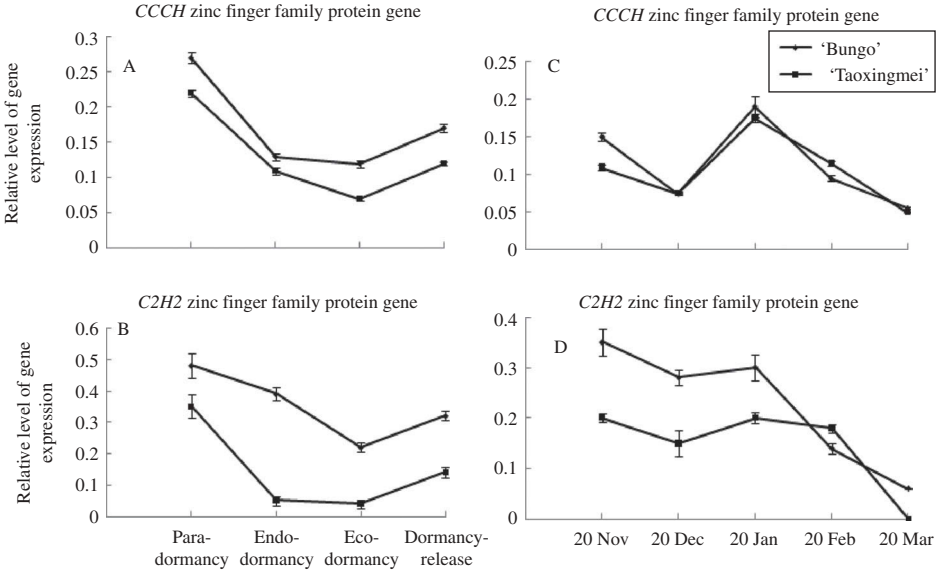


Figure 3. Changes in the relative levels of expression of two zinc-finger family protein genes *CCCH* (Panels A, C) and *C2H2* (Panels B, D) in flower buds (Panels A, B) and vegetative buds (Panels C, D) of two cultivars of Japanese apricot, 'Bungo' and 'Taoxingmei'. Para-, endo-, and eco-dormancy, and the dormancy-release stages coincided with sampling dates of 10 December 2012, 11 January, 8 February, and 26 February 2013 for 'Bungo' (high chilling requirement), and sampling dates of 9 November and 12 December 2012, and 23 January and 24 February 2013 for 'Taoxingmei' (low chilling requirement). As flower buds and vegetative buds had different sampling dates and criteria, the *x*-axes differ between Panels A, B and Panels C, D. Values are means ± SD (n = 3). Relative levels of gene expression at each stage of dormancy marked with an asterisk were significantly different between cultivars at $P \leq 0.05$ using Duncan's multiple range test.

Transcript levels of both zinc-finger family genes were lower in vegetative buds of 'Taoxingmei' than in 'Bungo' in November, December, and January, but not in February.

Expression pattern of the GA20ox gene in flower buds

Transcription of *GA20ox* was low during para-dormancy in flower buds of both cultivars (Figure 4A), and was up-regulated slightly during endo-dormancy, before becoming further up-regulated during eco-dormancy, and down-regulated slightly at the dormancy-release stage.

Levels of expression of *GA20ox* in flower buds of the low chilling requirement 'Taoxingmei' were higher than in 'Bungo' at the para-, endo-, and eco-dormancy, and dormancy-release stages.

Expression patterns of the GA20ox gene in vegetative buds and young leaves

Expression of *GA20ox* (Figure 4B) was lower on 20 December than on 20 November in vegetative buds of both 'Bungo' and 'Taoxingmei', but was up-regulated on 20 January, and up-regulated significantly on 20 February. However, transcript levels of *GA20ox* were much lower during the first 2 months of sampling than those on 20 January and 20 February. Transcript levels of *GA20ox* were significantly lower in young leaves of both

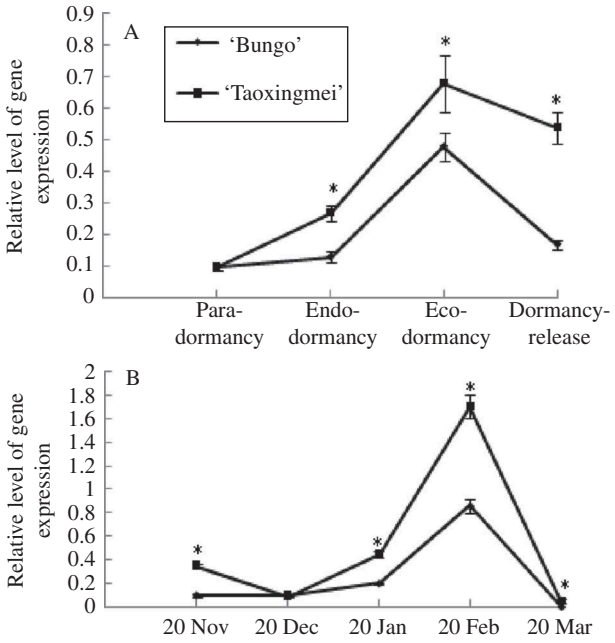


Figure 4. Changes in the relative levels of expression of the *GA20ox* gene in flower buds (Panel A), or vegetative buds and young leaves (Panel B) of two Japanese apricot cultivars, 'Bungo' and 'Taoxingmei'. Para-, endo-, and eco-dormancy, and the dormancy-release stages coincided with sampling dates of 10 December 2012, 11 January, 8 February, and 26 February 2013 for 'Bungo' (high chilling requirement), and sampling dates of 9 November and 12 December 2012, and 23 January and 24 February 2013 for 'Taoxingmei'(low chilling requirement). Values are means ± SD (n = 3). Relative levels of gene expression at each stage of dormancy marked with an asterisk were significantly different between cultivars at $P \leq 0.05$ using Duncan's multiple range test.

cultivars than in vegetative buds collected on 20 February.

Discussion

ABA and GA_3 are involved in bud dormancy in Japanese apricot

The first step towards establishing dormancy is the cessation of growth. Abscisic acid (ABA) is a plant hormone that is generally considered to be a potent inhibitor of growth. Saab et al. (1990) suggested that the accumulation of endogenous ABA inhibited shoot elongation directly under low water potential in maize. Endogenous ABA played a major role in both the cessation of development and the induction of dormancy (Walbot, 1978; Walton, 1980; Wareing & Bradbeer, 1978). ABA was involved in the onset and maintenance of dormancy in seeds (Fincher, 1989). In this study, ABA concentrations were high during para-dormancy in 'Bungo' and 'Taoxingmei'. This agreed with Rohde et al. (2002), who observed that ABA concentrations in apical buds increased significantly during short days. The biosynthesis of ABA in buds was further supported by expression of the poplar gene homologue of *VIVIPAROUS14*, which encodes a dioxygenase that is thought to regulate the first committed and rate-limiting step in ABA biosynthesis (Schwartz et al., 1997) in September and October. Moreover, Ruttink et al. (2007) reported that genes encoding enzymes involved in ABA biosynthesis were up-regulated after exposure to short days for 3–4 weeks. These observations indicate that ABA might act as an inhibitor of growth in flower buds during para-dormancy.

Although ABA concentrations declined during endo-dormancy compared to para-dormancy in both cultivars, ABA concentrations were higher during the first three stages of dormancy compared to the dormancy-release stage. González-Grandío et al. (2013) suggested that a correlation existed between ABA concentration and bud arrest in *A. thaliana*. Axillary buds of wild-type plants, which become dormant after a shade treatment, exhibited a significant increase in the global response of ABA-related genes, whereas axillary buds of the *A. thaliana brc1* (branched1) mutant, which continued to grow, showed reduced ABA-related responses. Therefore, a high concentration of ABA in dormant buds might play a prominent role in bud suppression (Chatfield et al., 2000) and in the inhibition of shoot growth (Lenoble et al., 2004).

During all four stages of dormancy, the concentrations of ABA in flower buds were lower during eco- and endo-dormancy than para-dormancy in both cultivars. This was similar to the findings of Wright (1975), who observed a decline in ABA concentrations during dormancy in several plant species.

One of the functions of GA_3 is to act as an antagonist to ABA and to promote germination (Gubler et al., 2005). In this study, concentrations of GA_3 increased significantly during eco-dormancy and dormancy-release compared to para- and endo-dormancy in flower buds of 'Bungo' and 'Taoxingmei'. Horvath et al. (2003) indicated that GAs played an important role in controlling the release of bud dormancy. Others have reported that GA concentrations might be regulated by the circadian clock, which responds to environmental signals such as light and temperature during dormancy (Penfield & Hall, 2009). Therefore, GA_3 might contribute to the regulation of dormancy in Japanese apricot, especially during the dormancy-release stage.

Concentrations of ABA and GA_3 differed significantly between 'Bungo' and 'Taoxingmei'. 'Taoxingmei', which has a low chilling requirement, had lower ABA concentrations and higher GA_3 concentrations than 'Bungo', which showed early bud burst.

In summary, bud dormancy in Japanese apricot has been shown to be related to changes in ABA and GA_3 concentrations during the four key stages of dormancy in both cultivars.

Zinc-finger family genes may be associated with bud dormancy in Japanese apricot

Previous studies have shown that several members of the zinc-finger gene family (*B-box, CCCH, C2H2*, and *C3HC4/RING*) were associated with dormancy-release in blackcurrant buds (Hedley et al., 2010). In 'Bungo' and 'Taoxingmei', the zinc-finger family gene *CCCH* was highly expressed in flower buds during para-dormancy, and in vegetative buds on 20 November. This suggested that zinc-finger protein genes might have a conserved function in the maintenance of dormancy. Similarly, Lee et al. (2012) reported that an *A. thaliana CCCH* zinc-finger protein gene, *AtC3H49*/*AtTZF3*, negatively regulated the growth rate of seedlings. They found that levels of expression of *AtC3H49*/*AtTZF3* and *AtC3H20*/*AtTZF2* were increased by ABA and several abiotic stresses, suggesting that proteins of the CCCH zinc-finger family might be involved in ABA and/or stress responses. Lin et al. (2011) suggested that the CCCH zinc-finger protein, AtTZF1, affected ABA-mediated plant growth. In both Japanese apricot cultivars, *CCCH* transcripts accumulated most in flower buds during para-dormancy, concomitant with the highest concentration of ABA.

In this study, the zinc-finger gene *C2H2* was significantly up-regulated during dormancy-release, compared to during eco-dormancy, in flower buds of both cultivars. The level of expression of *C2H2* was also high in vegetative buds, but was low in

young leaves of both cultivars. These results suggest that *C2H2* gene expression might be closely associated with the induction of dormancy-release in flower buds of Japanese apricot.

These data indicated that expression of *CCCH*- and *C2H2*-type zinc-finger genes might be correlated with ABA concentrations. Further analysis and biotechnological approaches will be required to reveal details of the relationship between the expression of zinc-finger family genes and ABA concentrations.

GA20ox might be associated with bud dormancy in Japanese apricot

In this study, the *GA20ox* gene was highly expressed in flower buds during the eco-dormancy to dormancy-release stages, and in vegetative buds on 20 February. *GA20ox* is a key gene associated with GA-signalling. Middleton et al. (2012) used mathematical modelling and experiments to explain how the *GA–GID1–DELLA* feedback loops interacted to control GA-signalling. They found that *GA20ox* played a significant role in modulating the level of endogenous GAs (such as GA_4) and generated overshoots after the removal of exogenous GAs. Moreover, transcriptional regulation of *GID1* and *DELLA* acted synergistically with *GA20ox* feedback. Ruttink et al. (2007) found that *GA20ox* was expressed at low levels after 3–4 weeks of short days. Therefore, the level of expression of *GA20ox*, which is closely associated with the concentration of GA_3, was low during the early stages of dormancy and high during the late stages of dormancy. This suggests that *GA20ox* might play a prominent role in dormancy by regulating GA signalling in Japanese apricot, especially that associated with the regulation of GA_3 concentrations.

Conclusions

In conclusion, dormancy in Japanese apricot buds was closely related to changes in the concentrations of ABA and GA_3. A high concentration of ABA during para-dormancy may inhibit the growth of flower buds and was associated with high levels of expression of *CCCH* and *C2H2*-type zinc-finger genes, whereas a high concentration of GA_3 may affect dormancy, especially at the dormancy-release stage, and was associated with high levels of expression of the *GA20ox* gene. Moreover, Japanese apricot genotypes with a low chilling requirement might possess lower ABA concentrations and higher GA_3 concentrations than Japanese apricot genotypes with a high chilling requirement, resulting in early dormancy-release.

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